

Contact Strength & Finger Power

An 8-week periodized finger-strength block for climbers

The Kinesiology of Climbing · Practice program · v1.0

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Read this first. This is an educational strength-and-conditioning program for **apparently healthy, screened adults** who choose to train at their own risk. It is **not** medical advice, physiotherapy, rehabilitation, or a plan for training around an injury. Before you start, complete the site's readiness screening ([legal/par-q.html](#)) and read the **Health & Fitness Disclaimer** ([legal/disclaimer.html](#)). If anything is flagged, get clearance from a physician or qualified professional first. See the full **Safety & Screening** section below — it is mandatory reading, not fine print.

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1. Overview — what this develops and why it transfers

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This is the classic **low-rep, high-force finger block**: the training that raises how much force your fingers and pulling chain can produce in a single, brief, near-maximal effort.

Three linked qualities are the target:

- **Contact strength** — the force you can express almost instantly when you catch or latch a hold. It is limited by how quickly and completely your muscles switch on.
- **Maximal finger force** — the ceiling of how hard your forearm flexors can pull through the finger tendons on a given edge.
- **Recruitment** — how fully your nervous system activates the available muscle. Early strength gains are dominated by this *neural* side rather than by muscle size, which is exactly what climbers want: **more force with very little added mass**, so your strength-to-weight ratio improves.

The transfer, in plain terms. Hard moves are limited by how much force you can produce *right now* on the hold in front of you. Train that ceiling up, and smaller holds become usable and bigger moves become stickable. Because the stimulus here is high-intensity and low-volume, most of the adaptation is neural — you get stronger without getting heavier. This is the same logic laid out in the site's *Strength ≠ size* and *Finger anatomy* articles: finger "strength" is really **forearm strength plus healthy tendons and pulleys**, and low-rep loading is the honest way to raise the force ceiling.

What this program is *not*. It is not an endurance or "anti-pump" program (see *Anaerobic Capacity for Long Routes* for that), not a first lifting program, and not a rehab plan. It will not fix a tweaky pulley — it assumes healthy, ready fingers.

2. Who this is for — prerequisites and the honest "not yet" gate

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This block is a good fit if you:

- Have **at least ~2 years of consistent climbing** and a training age that includes some structured finger loading (you have hung a board before, or bouldered hard on small holds for a season or more).
- Climb roughly **V4/6c+ (5.11+) or harder**, or can comfortably do controlled bodyweight hangs on a **20 mm edge** in a half-crimp for 10+ seconds.
- Can perform **at least 3–5 strict pull-ups** and have no current pain in fingers, elbows, or shoulders.
- Have **passed the readiness screening** and are apparently healthy.

This is *not* for you yet if any of the following are true:

- **You have climbed for less than about 1 year**, or have never trained fingers deliberately. Your tendons and pulleys need more time under progressive climbing load first. Start with general climbing volume and the free *Antagonist & Shoulder Resilience* and *Off-the-Wall Foundations* material. Muscle gets strong faster than tendon does; rushing this is the single most common way climbers pop a pulley.
- **You have a current or recent (within ~3 months) finger, pulley, elbow, or shoulder injury**, or any lingering finger pain. This program adds load; it is not a return-to-climbing protocol. See a physiotherapist or sports physician and get cleared first.
- **You are an adolescent or your growth plates may still be open**. This is a genuine safety issue, not a formality. In growing climbers the **epiphyseal growth plate at the base of the middle phalanx** is a recognised weak point, and **epiphyseal (growth-plate) stress fractures of the fingers are well documented in young climbers**, particularly with hard crimping and hangboarding. Growth plates typically close in the **mid-to-late teens to around age ~18–19, but timing varies widely between individuals**. If you are under 18, are unsure whether your growth plates have closed, or have any finger pain during growth, **do not hangboard maximally** — train fingers through climbing itself and seek guidance from a sports physician. When in doubt, wait. (*This is general education, not a diagnosis; a clinician can assess skeletal maturity.*)
- **You cannot yet hang your own bodyweight** on a ~20 mm edge with control, or you are new to resistance training generally. Build that base first.

Bottom line: finger strength is worth chasing, but the connective tissue that carries the load adapts in **months, not weeks**. The prerequisites above exist so the training stimulus lands on tissue that is ready for it.

3. Safety & screening (mandatory)

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Finger training carries **real, specific injury risk** — most commonly strains or tears of the **A2 and A4 finger pulleys**, and flexor tendon or growth-plate problems. Treat this section as part of the program, not a disclaimer.

Before you begin

1. **Complete the readiness screening** — the site's PAR-Q ([legal/par-q.html](#)). If it flags anything, get medical clearance before training.
2. **Read the Health & Fitness Disclaimer** ([legal/disclaimer.html](#)). You train voluntarily and at your own risk.
3. **Confirm you meet the prerequisites** in Section 2, including the growth-plate gate.

This is education, not individualized medical advice

This program is general strength-and-conditioning information for apparently healthy adults. It does **not** create a coach–client, physician–patient, or physiotherapist–patient relationship, does not diagnose or treat any condition, and is **not** a substitute for in-person assessment. For anything specific to your body, work with a qualified professional.

Pulley-injury warning signs — know these

Stop and back off if you notice any of the following. If they persist, see a physiotherapist or sports physician before loading fingers again:

- A **sharp pain or an audible/palpable "pop"** at the base or middle of a finger — this is the classic pulley-injury presentation and is a **stop-now, seek-assessment** event.
- **Swelling, tenderness, or a "bowstringing" sensation** (the tendon lifting away from the bone) along the front of a finger.
- **Pain that is sharp, localized to a joint or the palm-side of a finger**, as opposed to the diffuse muscular ache of a worked forearm.
- Finger pain that **worsens session to session**, or morning stiffness/pain that lingers.

Pain rules — the stop criteria

- **Never push through sharp pain.** Dull, symmetrical forearm fatigue is expected; sharp, localized, or one-sided finger pain is a stop signal.
- Use a simple traffic-light rule during and after sessions:
 - **Green (0–2/10, dull, diffuse):** proceed as planned.
 - **Amber (3/10, or any twinge that is localized but fades):** reduce load, switch to open-hand, cut volume, and reassess next session. Do not add load.
 - **Red (sharp pain, a pop, pain that lingers >24–48 h, or any warning sign above):** stop the finger work, do not "test" it, and if it does not settle quickly, get it assessed.
- **When in doubt, do less.** A missed session costs you nothing; a popped A2 costs months.

~~If you have any finger, elbow, or shoulder pain, a history of pulley injury, or any doubt about your readiness, consult a physiotherapist or sports-medicine physician before starting. Nothing here replaces individual clinical judgement.~~

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4. Equipment and substitutions

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You need a way to **grip a defined edge** and a way to **add or remove load**. That's it.

ITEM	PREFERRED	SENSIBLE SUBSTITUTIONS
Edge / hangboard	A hangboard with a flat 20 mm edge (the standard benchmark depth) plus a larger edge for warm-ups	A quality wooden edge/block hung from a rope or rig; a portable edge (e.g. a hangboard block) clipped to a pull-up bar
Add load	Weight belt or harness + weight plates/pins; a loading pin	Dumbbell held between the feet/knees; weight vest; backpack loaded with plates or water
Remove load (assistance)	Resistance band looped over the bar (feet or knee in the band) or a pulley/counterweight system	A stool/chair to take partial bodyweight through the legs (measured/repeatable), or a foot on a scale to quantify offloaded weight
Pull-ups	Pull-up bar + means to add weight	Rings; assisted-pull-up band
Timing	A repeater/interval timer app, or a coach calling time	Any clock with a second hand
Tracking	The site's built-in workout logger (Practice page)	A notebook or spreadsheet — log edge depth, added/removed load, and RPE every set

Notes on substitutions

- **No-hang / lifting-edge devices** (a portable edge attached to a loading pin, lifted from the floor) are an excellent and often *safer* substitute for max-hang work: load is easy to dial precisely and you can bail instantly. Anywhere this program prescribes a **max hang**, you may substitute a **no-hang lift** of matched intensity and grip type.
- If you **cannot yet add weight to a bodyweight hang**, that's fine — use **smaller edges** or **band assistance** to reach the target effort, and progress by removing assistance or reducing edge depth. See the assessment section.
- Avoid tiny, aggressive "micro" edges and pockets for this block unless you are advanced; the 20 mm edge is the workhorse because it loads the fingers hard while keeping the pulleys in a kinder position.

5. Assessment & benchmarking — establishing your baseline

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You cannot autoregulate load without a baseline. Do the assessment on a **fresh day** (not after hard climbing), fully warmed up (Section 8). Pick **one** protocol below and use it for the whole block so retests are comparable.

The benchmark grip and edge

- **Edge:** flat **20 mm** (record the exact depth you use).
- **Grip:** **half-crimp** (fingers ~90° at the big knuckle, thumb *off*) as the primary test grip. If half-crimp is uncomfortable, use **open-hand/open-drag** and record that.
- **Do not test in a full crimp.** (See the exercise library on full-crimp risk.)

Protocol A — Maximal Added Weight (MAW) (*preferred if you can add load*)

1. Warm up thoroughly.
2. Find the **most added weight you can hold for a controlled ~10-second hang** on the 20 mm edge in a half-crimp, with good form (shoulders engaged, no elbow lock, no shaking your way through it).
3. Work up in small jumps (e.g. +5 kg, then +2.5 kg) with **3+ minutes rest** between attempts. Stop when you can't hold a clean 10 s.
4. Record: **bodyweight + added weight = your effective hang load**. This added-weight figure is your **MAW**. Training percentages below are taken from MAW where noted, or set by RPE.

Protocol B — Max hang time (*if you can't/shouldn't add weight*)

1. Warm up thoroughly.
2. On the 20 mm edge (or the smallest edge you can hold safely) in a half-crimp, time your **maximum controlled hang to form failure** (stop the instant form breaks — no clawing).
3. Record the time. If bodyweight on 20 mm is easy (>20 s), use a smaller edge or add light load and record that setup.

Protocol C — Assisted (band/pulley) baseline (*if you can't yet hang bodyweight*)

1. Loop a band over the bar and put a foot/knee in it to offload weight.
2. Find the **least assistance** that lets you hold a clean 10 s hang on 20 mm.
3. Record the band (or measured kg offloaded). Progress by removing assistance.

Turning the baseline into training loads

- **Preparatory work (Weeks 1–2): ~RPE 6–7** — you finish each hang feeling you had ~3–4 seconds / a solid rep or two in reserve. In MAW terms this is roughly **70–80% of MAW**.
- **Max-strength work (Weeks 3–5): ~RPE 8** — ~1–2 in reserve; roughly **85–90% of MAW**.
- **Peak-force work (Weeks 6–7): ~RPE 9** — very hard but *never* to failure or to form breakdown; roughly **90–95% of MAW** on short holds.

• **Retest** at the end (Section 13) using the identical protocol.

CONTACT STRENGTH & FINGER POWER

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Honesty note: RPE for finger hangs is less precise than for barbell lifts, and MAW can drift a couple of kg day to day with sleep, skin, and warm-up. Treat the percentages as a starting map, then **let RPE and form be the final authority** each session.

6. How the block is built — the periodization, phase by phase

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This is an **8-week mesocycle** built on a simple, well-supported idea (see the *Periodization 101* article): apply a stress, let the body adapt, then evolve the stress so adaptation keeps coming — and **deload on purpose** so the fitness surfaces. Intensity climbs across the block while volume drops, which is the standard shape of a strength block.

WEEKS	PHASE	INTENSITY	"REPS" (HANGS)	REST	PRIMARY FOCUS	WHY THIS PHASE EXISTS
1–2	Preparation / grip tolerance	Moderate (RPE 6–7)	6–8	90 s	Re-groove technique, prime tendons & pulleys	Longer, submaximal loading rebuilds tissue tolerance and skill before heavy work — muscle adapts fast, connective tissue slowly, so we let it catch up first
3–5	Max strength / recruitment	High (RPE 8)	3–5	3 min	Recruitment & contact strength	Heavy, low-rep hangs drive the <i>neural</i> gains — fuller, faster muscle activation — that raise the force ceiling with minimal added mass
6–7	Peak force	Very high (RPE 9)	1–3	3–4 min	Maximal finger & pull force	Very short, very heavy efforts express the strength built in the prior phase; kept brief and well-rested to stay safe
8	Deload & retest	Low	reduced	—	Recover, then measure	Backing off lets accumulated fatigue drain so the adaptation shows up on the retest

The logic in one paragraph. Weeks 1–2 make your fingers *ready* to be loaded hard. Weeks 3–5 do the main work of teaching your nervous system to pull harder. Weeks 6–7 sharpen that into peak, near-instant force. Week 8 lets it all surface so you can measure a real gain and set up the next block. Each phase is deliberately built to set up the next — skipping the prep phase to "get to the heavy stuff" is how tissue gets ahead of itself.

Frequency: finger-intensive sessions are kept to **2 per week with ≥48 h between them**, plus one lighter pull-power/support session — three sessions total, which matches how much high-intensity finger loading most climbers recover from while still climbing.

7. Weekly structure & integrating with climbing

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Golden rule of sequencing: do high-intensity finger work when you are **fresh**, not at the end of a pump session.

Hard hangs go **early in a session and early in the week**, before or well-separated from hard climbing — never on trashed fingers.

Default week (3 training sessions + climbing)

DAY	SESSION	NOTES
Mon	Session A — Max fingers + weighted pull	Fresh day. Fingers first.
Tue	Rest or easy skills / mobility	No hard gripping
Wed	Session B — Pull-power + support/antagonist	Low finger load
Thu	Rest or easy climbing (volume, technique)	Keep it submaximal
Fri	Session C — Max fingers + contact strength on the wall	≥48 h after Session A
Sat	Climbing day (project / bouldering)	Your normal climbing
Sun	Rest	Full recovery

- ≥48 hours between the two heavy finger days (A and C). Never stack them.
- Put your **hardest climbing on Saturday**, away from the heavy finger days.
- Total hard-finger exposures per week (hangs + limit bouldering) should stay **moderate** — more is not better here; recovery is where fingers get stronger.

If you're short on time

Cut in this order, protecting the primary stimulus:

1. First cut **Session B** (support work) — do a shortened version on a climbing day.
2. Then trim the **weighted pull** volume in Sessions A/C.
3. **Never** cut the warm-up, and **never** cut the rest periods to save time — long rests are part of the strength stimulus, not padding.
4. A **two-session week** (A and C only) is a legitimate minimum-effective-dose week.

Managing total load & recovery

- Sleep and food drive tendon and neural adaptation more than any extra set does.
- If climbing volume is high that week, **reduce hang volume**, not intensity — keep the heavy stimulus, cut the number of sets.
- Skin and joints are part of the system: shredded tips or achy A2s mean **back off**.

8. The warm-up (non-negotiable)

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Never load fingers hard cold. Budget **15–20 minutes**. The goal is warm tissue, raised heart rate, and progressively loaded fingers — not fatigue.

Stage 1 — Pulse raiser (3–5 min). Light cardio: skipping, jogging, jumping jacks, or arm swings until you're slightly warm.

Stage 2 — Mobility & tissue prep (3–5 min). Wrist circles, finger flexion/extension, gentle finger-tendon glides, band pull-aparts, and shoulder circles. A few easy scapular pull-ups.

Stage 3 — Progressive finger loading (8–10 min). This is the important part. Do **3–5 easy hangs**, starting on a **large, comfortable edge** and progressing toward your working edge/grip, ramping load gradually:

SET	EDGE / GRIP	EFFORT	HOLD	REST
1	Large jug/edge, open hand	Very easy	10 s	60 s
2	Large edge, half-crimp	Easy	10 s	60 s
3	20 mm, half-crimp, bodyweight	Moderate	8–10 s	90 s
4	20 mm, + ~50% of today's added load	Brisk	7 s	90 s
5 (heavy days)	20 mm, + ~80% of working load	One "primer"	5 s	2–3 min

If **anything feels sharp** during the warm-up, stop and reassess — don't proceed to heavy work on a finger that's already complaining.

9. Week-by-week session prescriptions

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How to read these tables. "Hangs" are the working sets. Loads are set by **RPE first**, with the MAW percentage as a guide. Every session is preceded by the full warm-up. Rest times are minimums — take more if you need it. Keep every hang **7–10 s and clean**; stop the set the moment form breaks.

Grip: alternate or split your work between **half-crimp** and **open-hand** across the week to spread load kindly across the tissues (see the finger-anatomy rationale). Where a single grip is listed, the other can be used on the second finger day.

EXERCISE	EDGE / GRIP	SETS × HOLD	REST	TARGET	CUE
Max hang (submax)	20 mm, half-crimp	4 × 8–10 s	90 s	RPE 6–7 (~70–80% MAW)	Solid, controlled; 2–3 s in reserve
Max hang (submax)	20 mm, open-hand	3 × 8–10 s	90 s	RPE 6–7	Spread the load, no clawing
Weighted pull-up	Bar/rings	3 × 5	2–3 min	RPE 7	Full range, controlled

Session B (pull-power + support)

EXERCISE	SETS × REPS	REST	TARGET
Explosive pull-up (or lock-off holds)	4 × 3	2 min	Fast up, controlled down
Wrist-extensor eccentrics / reverse curl	3 × 12–15	60 s	Elbow prehab
Shoulder external rotation (band/cable)	3 × 12–15	60 s	Cuff health
Anti-rotation core (Paloof)	3 × 10/side	60 s	Body tension

Week 1 = learn the loads. Week 2 = same structure, nudge load up if Week 1 felt ≤ RPE 6.

EXERCISE	EDGE / GRIP	SETS × HOLD	REST	TARGET	CUE
Max hang	20 mm, half-crimp	4–5 × 7–10 s	3 min	RPE 8 (~85–90% MAW)	Hard but clean; 1–2 in reserve
Max hang	20 mm, open-hand	3 × 7–10 s	3 min	RPE 8	Keep it honest
Weighted pull-up	Bar/rings	4 × 3–5	3 min	RPE 8	Heavier than Phase 1

Session B (pull-power + support): as Phase 1, add load to the pull-power exercise.

Weekly progression: if the top set of max hangs finishes at ≤ **RPE 8 with clean form**, add **2.5–5% load** (or ~2.5 kg) next session. If it hit RPE 9+ or form slipped, **hold or reduce**. See Section 11.

EXERCISE	EDGE / GRIP	SETS × HOLD	REST	TARGET	CUE
Heavy max hang	20 mm, half-crimp	5 × 3–5 s	3–4 min	RPE 9 (~90–95% MAW)	Very heavy, very short, still clean
Max hang	20 mm, open-hand	2 × 5 s	3 min	RPE 8–9	Second grip, lower volume
Contact strength on the wall (Session C)	Wall	4–6 hard moves/attempts	2–3 min	High	Powerful latches on good holds; not a pump session
Weighted pull-up (low volume)	Bar/rings	3 × 2–3	3 min	RPE 8–9	Keep it snappy

Session B: reduce support volume slightly; keep pull-power crisp and low-rep.

Advanced option (Session A only, one grip): experienced climbers with healthy fingers may include a **small amount** of heavier, smaller-edge work here — but keep it minimal, open-hand or half-crimp, and stop at the first sign of a warning sign.

PHASE 4 — Deload & retest (Week 8)	
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DAYS	WHAT TO DO
Early week (2 sessions)	Warm-up + 2 × 5 s submax hangs (RPE 5–6) on 20 mm, one grip. Light pull-ups. Easy, technical climbing only. No max efforts.
Late week (fresh day)	Retest using your exact baseline protocol (Section 5). Then rest.

Deload is not optional and not a sign of weakness — it's when the adaptation surfaces.

10. Exercise library

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Each entry: what it's for, setup, grips, tempo, common errors, and how to scale.

Max hang (the core exercise)

- **Develops:** maximal finger force and recruitment on a fixed edge.
- **Setup:** warm up fully. Grip the edge, take a **strong active-shoulder position** (shoulders down and slightly engaged — *not* hanging passively off the joints), soft elbows. Hang for the prescribed **7–10 s**, then step or lower off under control.
- **Grips:**
 - **Open-hand / open-drag** — fingers more extended; kindest to the pulleys; use for a large share of volume.
 - **Half-crimp** — fingers ~90° at the big knuckle, thumb *off*; the key trainable position for climbing and the primary benchmark grip.
 - **Full crimp (fingers bent sharply, thumb clamped over)** — **trained cautiously, if at all**. The full crimp places the **highest load on the A2 pulley** and is the position most associated with pulley injury. This program **does not** prescribe max full-crimp hangs. Advanced, healthy, experienced climbers may maintain some full-crimp exposure through climbing, but do not chase max loads in it here.
- **Tempo:** static hold; no bouncing, no re-gripping mid-hang.
- **Common errors:** hanging passively off the shoulders; letting the fingers drag/claw open toward a crimp under fatigue; holding too long and grinding out ugly seconds (stop at form failure); testing on cold fingers.
- **Regressions:** larger edge; band/foot assistance; shorter holds; reduce load.
- **Progressions:** add load; reduce edge depth (advanced); progress from Phase 1 → 3 intensities.

No-hang / edge-lift (*equal-status alternative to max hangs*)

- **Develops:** same as max hangs, with **precise, instantly-bailable load**.
- **Setup:** attach an edge to a loading pin/handle; stand tall, brace, and lift the load off the floor through the fingers only, matching the grip and hold time of the max hang it replaces.
- **Why use it:** easy to load exactly, easy to drop, low whole-body demand — often the safer choice. Substitute anywhere max hangs appear.

Weighted pull-up

- **Develops:** maximal pulling force in the arms and back that backs up the fingers.
- **Setup:** add load via belt/vest/dumbbell. Full hang to chin-over; **control the descent**.
- **Grips:** overhand on a bar or neutral on rings (rings are kinder to elbows).
- **Tempo:** intentful up, ~2–3 s down. In peak phase, keep it snappy and low-rep.
- **Errors:** kipping, partial range, shrugging into the ears at the top.
- **Regressions:** bodyweight or band-assisted. **Progressions:** add load; slow eccentric.

- **Develops:** rate of force development — turning strength into fast, forceful pulling.
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- **Setup (explosive):** from a dead hang, pull as fast as possible to chest height; lower under control. Quality over quantity — stop when speed drops.
- **Setup (lock-off):** hold a static bent-arm position (e.g. 90°) for 3–5 s.
- **Errors:** grinding slow reps and calling them explosive; sacrificing shoulder position.
- **Scale:** band-assist explosive reps; shallower lock-off angle to start.

Support / antagonist work (elbow & shoulder health)

- **Wrist-extensor eccentrics / reverse curls** — for "climber's elbow" prehab.
- **Shoulder external rotation (band/cable)** — rotator-cuff health.
- **Anti-rotation core (Pallof press)** — trunk tension that transfers to the wall.
- These balance the pulling-dominant demands of climbing; keep them in even when time is tight. (See *Antagonist & Shoulder Resilience* for the full rationale.)

Contact strength on the wall (Phase 3)

- **Develops:** expressing finger force dynamically — catching and latching under speed.
- **Setup:** on a bouldering wall, do **short, powerful sequences on good-to-moderate holds** with full rest between attempts. Focus on **snappy, precise latches**, not volume and not tiny holds. This is a strength stimulus, not a pump session — if you're getting pumped, you're doing endurance by accident.

11. Autoregulation, progression & deload logic

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Numbers on a page can't feel your fingers. Autoregulation means **adjusting the plan to how you actually show up**.

The daily readiness gate (before every finger session)

Run through this in the warm-up:

- **Any red-flag finger pain or a recent pop?** → No max work. Reassess / seek advice.
- **Slept badly, sick, skin trashed, or fingers achy from the last session?** → Drop to the lower end of the RPE range and cut a set or two. Keep intensity honest but volume small.
- **Warm-up primer hang feels heavier than usual at the same load?** → That's your true strength for today; **autoregulate load down** to hit the target RPE, don't force the planned kilos.

Progressing load (the strength-first rule)

Because the goal is **strength**, progress **intensity before volume**:

1. When the **top working set finishes at or below the target RPE with clean form**, add ~2.5–5% **load** (about 2.5 kg) next session, *or* drop edge depth slightly (advanced).
2. If you hit the top of the RPE range but form was clean, **repeat the same load** next time until it drifts down to the target — then progress.
3. If form broke, you hit RPE 9+ when 8 was the target, or pain appeared, **reduce load**.
4. **Do not** chase progression by piling on extra sets. More volume is not the lever here.

When to deload early (don't wait for Week 8 if you see these)

- Two sessions in a row where planned loads feel far harder than their target RPE.
- Persistent finger, elbow, or shoulder achiness that doesn't clear between sessions.
- Sleep/mood/motivation tanking, or plateaued/declining loads.

A mid-block deload = take 3–5 days of easy climbing and no max hangs, then resume where you left off. **Backing off on purpose is part of the plan.**

12. Transfer to the wall

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Strength you can't use is wasted. To convert this block into climbing:

- **Keep climbing through the block.** This is a *supplement* to climbing, not a replacement. The wall is where finger force becomes climbing skill.
- **Sequence smart:** hard hangs when fresh, hard climbing on separate days (Section 7).
- **In the peak phase, spend time on your project or on hard, powerful boulders.** New force needs to be rehearsed in real movement — latching, catching, and holding under body tension — for the gym numbers to show up as sends.
- **Expect the payoff to surface after the deload,** when fatigue drops. Judge the block by your climbing in the *following* weeks, not mid-block when you're carrying fatigue.
- **Don't out-climb your tissue.** Feeling strong is exactly when climbers latch a tiny crimp too enthusiastically and get hurt. Ramp on-the-wall crimping as patiently as you ramped the hangs.

13. Retest & what to do next block

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Retest (end of Week 8, fresh day)

Repeat your **exact** baseline protocol from Section 5 — same edge, same grip, same warm-up, rested. Record the new number and compare.

- **MAW protocol:** compare added weight for a clean 10 s hang.
- **Max-time protocol:** compare hang time on the same edge.
- **Assisted protocol:** compare assistance needed.

A meaningful gain over 8 weeks is realistic but modest — think **a few kg of added load** or a few seconds — and it's driven mostly by better recruitment. **Don't measure success only by the retest;** how your climbing feels over the next few weeks matters more.

What to do next

- **If you gained and feel good:** take a lighter **transition week** (easy climbing, minimal hangs), then either (a) run a **power / power-endurance block** to express the new strength on longer sequences, or (b) repeat this block from a new, higher baseline after 3–4 weeks of general climbing.
- **If you stalled:** you likely need either more recovery (were you deloading enough?) or a change of stimulus. Rotate to a different quality for a block — finger strength responds well to being left alone and revisited, not ground on continuously.
- **If anything got tweaky:** stop finger-max work, let it settle, and get assessed before loading heavy again. Consistency over months beats any single hard block.
- **Don't run max finger blocks back-to-back indefinitely.** Tendons and pulleys reward patience; alternate hard finger blocks with other qualities across the year.

14. FAQ & troubleshooting

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My fingers feel "tweaky" mid-block — what do I do? Treat it as amber/red (Section 3). Stop max work on that finger, switch to open-hand at much lighter loads or take finger-rest days, and don't "test" it. If a sharp pain, a pop, swelling, or bowstringing appeared, that's a stop-and-get-assessed event. A few missed sessions cost nothing; a pulley tear costs months.

I've plateaued — the loads won't go up. First check recovery: sleep, food, and whether you're actually deloading. Then check honesty of intensity (are you truly at the target RPE, or grinding ugly reps?). If both are fine, you may simply need a change of stimulus — finish the block, take a transition week, and rotate to a different quality. Strength rarely climbs in a straight line.

Can I train fingers more than twice a week to speed things up? No — that's the fastest route to an overuse injury here. Two heavy finger days plus climbing is plenty of high-intensity finger exposure for most people. Recovery is the adaptation.

Should I use a full crimp? It feels stronger. Not for max hangs in this program. The full crimp loads the A2 pulley the hardest and is the grip most linked to pulley injury. Train half-crimp and open-hand; let any full-crimp exposure come from normal climbing.

I'm travelling / stuck without a hangboard for a week. Take it as a light week. A portable edge or no-hang device covers most travel; if you have nothing, do bodyweight climbing-specific work, keep antagonist/support work going with bands, and resume the plan when you're back. A missed week won't undo the block.

My skin is destroyed. Back off. You can't train fingers hard through split tips, and pushing on shredded skin changes your grip mechanics in ways that invite injury. Rest, file, moisturise, and return when the skin is intact.

I'm not sure my growth plates are closed. Then don't hangboard maximally, and see Section 2 and a sports physician. This is a real safety issue for younger climbers, not a formality.

How long until I see it in my climbing? Often after the deload, and over the following few weeks, as fatigue clears and you rehearse the strength on the wall. Judge the block by your climbing in the month *after* it ends.

15. References & further reading

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These are foundational, widely-cited works in climbing science, sports medicine, and resistance training, listed by author/year and topic. Summaries throughout this program are simplified for a general audience. The principles below are high-consensus; a site owner who wishes to publish a formal bibliography can add full titles and page details from the originals — but nothing here depends on a specific edition or DOI.

Finger-strength & hangboard training

- **López-Rivera, E. & González-Badillo, J. J. (2012; and follow-up work).** Comparisons of maximal-load ("max added weight") hangboard training against intermittent/repeater methods in sport climbers — the empirical basis for the max-hang, low-rep approach used here.
- **Medernach, J. P., Kleinöder, H., & colleagues.** Fingerboard-specific training increases maximal finger strength in climbers — support for dedicated hangboard blocks over climbing alone.
- **Levernier, G., Laffaye, G., and related work.** Maximal finger force and rate of force development distinguish higher- from lower-performing climbers — why "contact strength" transfers to harder climbing.

Pulley & crimp biomechanics; injury

- **Schöffl, V., Hochholzer, T., and colleagues.** The standard body of work on finger **pulley injuries** in rock climbers — mechanism (crimp loading), grading, diagnosis, and management. Basis for the warning signs and stop rules in Section 3.
- **Vigouroux, L., Quaine, F., and colleagues.** Biomechanical modelling of flexor-tendon and **pulley loading in the crimp vs. open grip** — why the full crimp places the highest load on the A2 pulley. Basis for the open-hand bias and full-crimp caution.
- **Schöffl, I., Schöffl, V., and colleagues.** **Epiphyseal (growth-plate) stress fractures** of the finger in adolescent climbers — basis for the open-physis / under-18 gate in Section 2.

Tendon adaptation, progression & periodization (foundational principles)

- **Magnusson, S. P., Kjaer, M., and colleagues.** Tendon and collagen adaptation to mechanical loading — connective tissue remodels over **months**, more slowly than muscle or neural drive. Basis for the "let the tissue catch up" prep phase and the patience caveats.
- **American College of Sports Medicine — Position Stand (2009).** Progression models in resistance training for healthy adults — general overload, intensity-before-volume, and recovery principles.
- **Kraemer, W. J. & Ratamess, N. A. (2004).** Fundamentals of resistance training: progression and exercise prescription — dose-response and how intensity/volume are manipulated across a block.
- **Issurin, V. B. (2010).** Periodization principles (including block periodization) — the "apply a stress, adapt, then evolve the stimulus, and deload on purpose" logic of Section 6.

Related free reading on the site: *The climber's fingers: pulleys, tendons & staying healthy* ([articles/finger-anatomy.html](#)), *Periodization 101* ([articles/periodization.html](#)), *Strength ≠ size* ([articles/strength-to-weight.html](#)).

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